Readings

Sundaram: Chapter 4

Interior Optima

Let $f : S \to \mathcal{R}$, where $S \subseteq \mathcal{R}^L$. **We want to maximize f over S.**

Important: Minimizing f is the same as maximizing $-f$.

Interior Local Maximum

Instead of finding global maximum, we start by finding **interior local maxima** and use them to get to global maximum.

$\boldsymbol{x}^\star$ is called an **interior local maximum** if
(i) $\boldsymbol{x}^\star$ is an interior point of S
(ii) There exists $\epsilon > 0$ such that $f(\boldsymbol{x}^\star) \geq f(\boldsymbol{x})$ for all $\boldsymbol{x} \in B_\epsilon(\boldsymbol{x}^\star) \subseteq S$.
(if the inequality is strict $\boldsymbol{x}^\star$ is called strict interior local maximum)

Caution: At this moment we are excluding points that are not interior points of S. But there could be maximum on the boundaries. We have to identify them separately.

First order necessary condition

$\boldsymbol{x}^\star$ is interior local maximum (or minimum) and f is differentiable at $\boldsymbol{x}^\star$ then

$$\nabla f(\boldsymbol{x}^\star) = 0$$

Caution This is not a sufficient condition.
Example: $S = [-1, 1]$, $f(x) = x^3$. There is a critical point $x = 0$ but it is neither interior local minimum nor interior local maximum, $f(\epsilon) > f(0)$ and $f(-\epsilon) < f(0)$ for all $\epsilon > 0$.

Second Order Conditions

$f \in \mathcal{C}^2$ (that is second order partials are continuous)

1. $\boldsymbol{x}^\star$ is interior local maximum (resp. minimum) $\Rightarrow D^2 f(\boldsymbol{x}^\star)$ is negative semi definite (resp. positive semi definite)

$$\boldsymbol{z}^T \left[D^2 f(\boldsymbol{x}^\star) \right] \boldsymbol{z} \leq 0 \text{ (resp. } \geq 0) \text{ for all } \boldsymbol{z} \in \mathcal{R}^L$$

2. $\boldsymbol{x}^\star$ is a critical point ($\nabla f(\boldsymbol{x}^*) = 0$) and $D^2 f(\boldsymbol{x}^\star)$ is negative definite (resp. positive definite) $\Rightarrow \boldsymbol{x}^\star$ is a strict interior local maximum (resp. strict local minimum).

$$\boldsymbol{z}^T \left[D^2 f(\boldsymbol{x}^\star) \right] \boldsymbol{z} < 0 \text{ (resp. } > 0) \text{ for all } \boldsymbol{z} \in \mathcal{R}^L$$

Caution: Mind the Gap Between Necessary and Sufficient Conditions

Example: $S = \mathcal{R}^2$, $f(x_1, x_2) = x_1^3 - 3x_1x_2^2$. Only critical point is $\boldsymbol{x}^\star = (0, 0)$ and $f(0, 0) = 0$

$$D^2 f(\boldsymbol{x}^\star) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

which is both positive semi definite and negative semi definite. However $x^\star$ is neither local min nor local max. For $\epsilon > 0$, $f(\epsilon, 0) > 0$ and $f(-\epsilon, 0) < 0$.

Identifying Local Interior Maximum/ Minimum Using Second Order Condition

- If the Hessian matrix at a critical point is negative definite (resp. positive definite) then the critical point is interior local maximum (resp. interior local minimum)
- If the Hessian matrix at a critical point is negative semi definite (resp. positive semi definite) then the critical point may or may not be interior local maximum (resp. interior local minimum). We need to find our own way to check it.
- If the Hessian matrix at a critical point is none of the above then the critical point is neither interior local maximum nor interior local minimum.

Global maximum

$S = \{x \mid x \geq -2\}, f(x) = 2x^3 - 3x^2$.

Critical points are $x^\star = 0, 1$. Second order conditions imply 0 is a strict interior local maximum and 1 is strict interior local minimum. $f(0) = 0$ and $f(1) = -1$.

But the global minimum is the boundary point $x = -2$
$(f(-2) = -28)$
Global maximum does not exist, f is increasing in x and has no upper bound.

Caution:

1. Don't forget the boundary points.
2. You may find interior local maximum/minimum but that does not imply that global maximum/minimum exists.

Identifying Global Maximum/ Minimum

- Step 1: Use Weierstrass' Theorem or some other argument to prove that global maximum/minimum exists.
- Step 2: Find all critical points
- Step 3: Classify all critical points using second order condition
- Step 4: Compare functional values at all interior local maximum (resp. local minimum) and **boundary points** to find global maximum (resp. minimum)

Proof: First Order Condition

Suppose $\boldsymbol{x}^*$ is an interior local maximum (same proof for minimum)

We shall show by contradiction that $\nabla f(\boldsymbol{x}^*) = 0$.

Suppose $\nabla f(\boldsymbol{x}^*) \neq 0$. Then we can find $\boldsymbol{z} \in \mathcal{R}^L$ such that $\nabla f(\boldsymbol{x}^*) \cdot \boldsymbol{z} > 0$

Take $t > 0$ a scalar such that $(\boldsymbol{x}^\star + t\boldsymbol{z})$ belongs to the neighbourhood of $\boldsymbol{x}^*$ where $\boldsymbol{x}^*$ is the local interior maximum. Hence $[f(\boldsymbol{x}^* + t\boldsymbol{z}) - f(\boldsymbol{x}^*)] \leq 0$

Now, take a sequence of positive $\{t_n\}_{n=1}^{\infty}$ which is in the above neighbourhood and converges to 0

$$\lim_{t \to 0} \frac{f(\boldsymbol{x}^* + t\boldsymbol{z}) - f(\boldsymbol{x}^*)}{t} = D_{\boldsymbol{z}} f(\boldsymbol{x}^*) = \boldsymbol{z} \cdot \nabla f(\boldsymbol{x}^*) > 0$$

Therefore we can find t_n such that $[f(\boldsymbol{x}^* + t_n\boldsymbol{z}) - f(\boldsymbol{x}^*)] > 0$ which is a contradiction

Proof: Second Order Conditions

Proof: $L = 1$

Part 2: Take an interior point $x^\star$ such that $f'(x^\star) = 0$ and $f''(x^\star) < 0$ (since it is a scalar).

By continuity, there is a neighbourhood around $x^\star$ such that for all $x \in B_\epsilon(x^\star)$, $f''(x) < 0$.

$\Rightarrow f'$ is strictly decreasing in $B_r(x^\star)$. Since $f'(x^\star) = 0$

(i) $f'(z) > 0$ for $z < x^\star \Rightarrow f(x^\star) > f(z)$ for all $z < x^\star$

(ii) $f'(z) < 0$ for $z > x^\star \Rightarrow f(x^\star) > f(z)$ for all $z > x^\star$

Combining (i) and (ii), we obtain, $f(x^\star) > f(z)$ for all $z \in B_r(x^\star)$. Hence $x^\star$ is a strict interior local maximum.

Part 1: x^* is an interior local maximum but suppose $f''(x^\star) > 0$. But by Part 2, x^* would be strict interior local minimum. Hence a contradiction.

Proof: Second Order Conditions

Proof: $L > 1$. We shall only prove part 1. $\boldsymbol{x}^*$ is an interior local maximum.

Pick any $\boldsymbol{z} \in \mathcal{R}^L$, $\boldsymbol{z} \neq 0$
Define $\phi(t) = \boldsymbol{x}^\star + t\boldsymbol{z}$ and $g(t) = f(\phi(t))$.

$g : \mathcal{R} \to \mathcal{R}$ is composition of two functions: $f : \mathcal{R}^L \to \mathcal{R}$ and $\phi : \mathcal{R} \to \mathcal{R}^L$

$g(0) = f(\phi(0)) = f(\boldsymbol{x}^\star)$

$g(0) > g(t)$ for all t such that $(\boldsymbol{x}^\star + t\boldsymbol{z})$ belongs to the neighbourhood where $\boldsymbol{x}^\star$ is the interior local maximum.

Hence g has local maximum at $0 \Rightarrow g''(0) \leq 0$.

By applying chain rule twice, we get $g''(0) = \boldsymbol{z}D^2 f(\boldsymbol{x}^\star)\boldsymbol{z}$.

Thus $\boldsymbol{z}D^2 f(\boldsymbol{x}^\star)\boldsymbol{z} \leq 0$ for all $\boldsymbol{z} \in \mathcal{R}^l$. Hence $D^2 f(\boldsymbol{x}^\star)$ is negative semi definite.